Methods of Mathematical Physics A
Assignment 3

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1

1.1

1.1.1

$$\begin{aligned}\int_0^{2+\mathrm{i}} \operatorname{Re} z \,\mathrm{d}z &= \int_0^2 \operatorname{Re} z \,\mathrm{d}z + \int_2^{2+\mathrm{i}} \operatorname{Re} z \,\mathrm{d}z \\ &= \int_0^2 x\mathrm{d}x + \int_0^1 2\mathrm{i}\mathrm{d}y \\ &= \frac{1}{2} \times 2^2 + 2\mathrm{i} \\ &= 2 + 2\mathrm{i}.\end{aligned}$$

1.1.2

$$\begin{aligned}\int_0^{2+\mathrm{i}} \operatorname{Re} z \,\mathrm{d}z &= (2+\mathrm{i}) \int_0^1 2t \,\mathrm{d}t \\ &= 2 + \mathrm{i}.\end{aligned}$$

1.2

1.2.1

$$\begin{aligned}\int_C \frac{\mathrm{d}z}{\sqrt{z}} &= \int_0^\pi \mathrm{i}\,\mathrm{e}^{\mathrm{i}\frac{\theta}{2}} \mathrm{d}\theta \\ &= 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\Big|_0^\pi \\ &= 2\mathrm{e}^{\mathrm{i}\frac{\pi}{2}} - 2 \\ &= 2(\mathrm{i} - 1).\end{aligned}$$

1.2.2

$$
\begin{aligned}
\int_C \frac{\mathrm{d}z}{\sqrt{z}} &= \int_0^{-\pi} \mathrm{i}\,\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\mathrm{d}\theta \\
&= 2\mathrm{e}^{\mathrm{i}\frac{\theta}{2}}\Big|_0^{-\pi} \\
&= 2\mathrm{e}^{-\mathrm{i}\frac{\pi}{2}} - 2 \\
&= -2(1+\mathrm{i}).
\end{aligned}
$$

2

2.1

2.1.1

$\frac{\sin\frac{\pi z}{4}}{z^2-1}$ 在 $C:|z|=\frac{1}{2}$ 包围的单连通区域内解析，故由 Cauchy 定理，

$$
\oint_C \frac{1}{z^2-1}\sin\frac{\pi z}{4}\,\mathrm{d}z = 0.
$$

2.1.2

记

$$
f(z) = \frac{\sin\frac{\pi z}{4}}{z+1},
$$

$f(z)$ 在 $C:|z-1|=1$ 包围的单连通区域内解析，由 Cauchy 积分公式

$$
f(a) = \frac{1}{2\pi\mathrm{i}}\oint_C \frac{f(z)}{z-a}\,\mathrm{d}z,
$$

得

$$
\begin{aligned}
\oint_C \frac{1}{z^2-1}\sin\frac{\pi z}{4}\,\mathrm{d}z &= \oint_C \frac{f(z)}{z-1}\,\mathrm{d}z \\
&= 2\pi\mathrm{i}f(1) \\
&= \mathrm{i}\frac{\sqrt{2}}{2}\pi.
\end{aligned}
$$

2.1.3

用割线 $[-3\mathrm{i}, 3\mathrm{i}]$ 将积分路径切为两个半圆，记

$$
f(z) = \frac{\sin\frac{\pi z}{4}}{z+1}, \quad g(z) = \frac{\sin\frac{\pi z}{4}}{z-1},
$$

则由 Cauchy 积分公式得

$$\begin{aligned}\oint_{C_1}\frac{f(z)}{z-1}\,\mathrm{d}z &= 2\pi\mathrm{i}f(1)\\ &= \mathrm{i}\frac{\sqrt{2}}{2}\pi,\\ \oint_{C_2}\frac{g(z)}{z+1}\,\mathrm{d}z &= 2\pi\mathrm{i}g(-1)\\ &= \mathrm{i}\frac{\sqrt{2}}{2}\pi,\end{aligned}$$

故

$$\begin{aligned}\oint_C\frac{1}{z^2-1}\sin\frac{\pi z}{4}\,\mathrm{d}z &= \oint_{C_1}\frac{f(z)}{z-1}\,\mathrm{d}z+\oint_{C_2}\frac{g(z)}{z+1}\,\mathrm{d}z\\ &= \mathrm{i}\sqrt{2}\pi.\end{aligned}$$

2.1.4

沿虚轴用割线将积分路径切为两个半圆，记

$$f(z)=\frac{\sin\frac{\pi z}{4}}{z+1},\quad g(z)=\frac{\sin\frac{\pi z}{4}}{z-1},$$

结果同上一问，两条路径各包含一个奇点，由 Cauchy 积分公式得

$$\begin{aligned}\oint_C\frac{1}{z^2-1}\sin\frac{\pi z}{4}\,\mathrm{d}z &= \oint_{C_1}\frac{f(z)}{z-1}\,\mathrm{d}z+\oint_{C_2}\frac{g(z)}{z+1}\,\mathrm{d}z\\ &= \mathrm{i}\sqrt{2}\pi.\end{aligned}$$

2.2

2.2.1

记

$$f(z)=\frac{\mathrm{e}^{\mathrm{i}z}}{z+\mathrm{i}},$$

$f(z)$ 在路径 $C:|z-\mathrm{i}|=1$ 包围的单连通区域内解析，故由 Cauchy 积分公式得

$$\begin{aligned}\oint_C\frac{1}{z^2+1}\mathrm{e}^{\mathrm{i}z}\,\mathrm{d}z &= \oint_C\frac{f(z)}{z-\mathrm{i}}\,\mathrm{d}z\\ &= 2\pi\mathrm{i}f(\mathrm{i})\\ &= \frac{\pi}{\mathrm{e}}.\end{aligned}$$

2.2.2

用割线 $[-2,2]$ 切割积分路径，记

$$f(z)=\frac{\mathrm{e}^{\mathrm{i}z}}{z+\mathrm{i}},\quad g(z)=\frac{\mathrm{e}^{\mathrm{i}z}}{z-\mathrm{i}},$$

由 Cauchy 积分公式得

$$\begin{aligned}\oint_C \frac{\mathrm{e}^{\mathrm{i}z}}{z^2+1}\,\mathrm{d}z &= \oint_{C_1} \frac{f(z)}{z-\mathrm{i}}\,\mathrm{d}z + \oint_{C_2} \frac{g(z)}{z+\mathrm{i}}\,\mathrm{d}z \\ &= 2\pi\mathrm{i}f(\mathrm{i}) + 2\pi\mathrm{i}g(-\mathrm{i}) \\ &= \frac{\pi}{\mathrm{e}} - \pi\mathrm{e}.\end{aligned}$$

2.2.3

用割线 $[-1,1]$ 切割积分路径，记

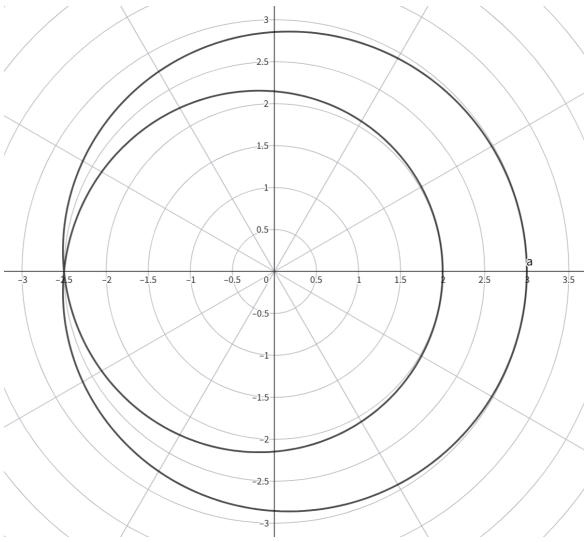
$$f(z) = \frac{\mathrm{e}^{\mathrm{i}z}}{z+\mathrm{i}}, \quad g(z) = \frac{\mathrm{e}^{\mathrm{i}z}}{z-\mathrm{i}},$$

同上一问，两个半椭圆连通区域各含一个奇点，由 Cauchy 积分公式得

$$\begin{aligned}\oint_C \frac{\mathrm{e}^{\mathrm{i}z}}{z^2+1}\,\mathrm{d}z &= \oint_{C_1} \frac{f(z)}{z-\mathrm{i}}\,\mathrm{d}z + \oint_{C_2} \frac{g(z)}{z+\mathrm{i}}\,\mathrm{d}z \\ &= 2\pi\mathrm{i}f(\mathrm{i}) + 2\pi\mathrm{i}g(-\mathrm{i}) \\ &= \frac{\pi}{\mathrm{e}} - \pi\mathrm{e}.\end{aligned}$$

2.2.4

作图：



积分路径可以分成两条闭合曲线，每条闭合曲线都包含了两个奇点. 对每个闭合曲线，用割线切割，再应用 Cauchy 积分公式得

$$\begin{aligned}\oint_C \frac{\mathrm{e}^{\mathrm{i}z}}{z^2+1}\,\mathrm{d}z &= \oint_{C_a} \frac{\mathrm{e}^{\mathrm{i}z}}{z^2+1}\,\mathrm{d}z + \oint_{C_b} \frac{\mathrm{e}^{\mathrm{i}z}}{z^2+1}\,\mathrm{d}z \\ &= \oint_{C_{a1}} \frac{f(z)}{z-\mathrm{i}}\,\mathrm{d}z + \oint_{C_{a2}} \frac{g(z)}{z+\mathrm{i}}\,\mathrm{d}z + \oint_{C_{b1}} \frac{f(z)}{z-\mathrm{i}}\,\mathrm{d}z + \oint_{C_{b2}} \frac{g(z)}{z+\mathrm{i}}\,\mathrm{d}z \\ &= 2\times 2\pi\mathrm{i}f(\mathrm{i}) + 2\times 2\pi\mathrm{i}g(-\mathrm{i}) \\ &= 2\left(\frac{\pi}{\mathrm{e}} - \pi\mathrm{e}\right).\end{aligned}$$

3

Proof. 记积分

$$
\begin{aligned}
I &= \int_0^{\pi} \mathrm{e}^{\cos\theta} \cos\left(\sin\theta\right) \mathrm{d}\theta \\
&= \frac{1}{2} \times \int_0^{\pi} \left(\mathrm{e}^{\cos\theta + \mathrm{i}\sin\theta} + \mathrm{e}^{\cos\theta - \mathrm{i}\sin\theta}\right) \mathrm{d}\theta \\
&= \frac{1}{2} \times \int_0^{\pi} \left(\mathrm{e}^{\mathrm{e}^{\mathrm{i}\theta}} + \mathrm{e}^{\mathrm{e}^{-\mathrm{i}\theta}}\right) \mathrm{d}\theta,
\end{aligned}
$$

记

$$
w = \mathrm{e}^{\mathrm{i}\theta},\ u = \mathrm{e}^{-\mathrm{i}\theta},
$$

则

$$
\begin{aligned}
I &= -\frac{\mathrm{i}}{2} \times \left(\int_0^{\mathrm{e}^{\mathrm{i}\pi}} \frac{1}{w}\mathrm{e}^{w}\mathrm{d}w - \int_0^{\mathrm{e}^{-\mathrm{i}\pi}} \frac{1}{u}\mathrm{e}^{u}\mathrm{d}u\right) \\
&= -\frac{\mathrm{i}}{2} \times \int_{\mathrm{e}^{-\mathrm{i}\pi}}^{\mathrm{e}^{\mathrm{i}\pi}} \frac{1}{w}\mathrm{e}^{w}\mathrm{d}w.
\end{aligned}
$$

计算

$$
\oint_{|z|=1} \frac{\mathrm{e}^{z}}{z}\mathrm{d}z,
$$

由 Cauchy 积分公式

$$
f\left(a\right) = \frac{1}{2\pi\mathrm{i}} \oint_C \frac{f\left(z\right)}{z-a}\mathrm{d}z,
$$

于是

$$
\begin{aligned}
\oint_{|z|=1} \frac{\mathrm{e}^{z}}{z}\mathrm{d}z &= 2\pi\mathrm{i}\mathrm{e}^{0} \\
&= 2\pi\mathrm{i}.
\end{aligned}
$$

代入原积分得

$$
\begin{aligned}
I &= -\frac{\mathrm{i}}{2} \times \int_{\mathrm{e}^{-\mathrm{i}\pi}}^{\mathrm{e}^{\mathrm{i}\pi}} \frac{1}{w}\mathrm{e}^{w}\mathrm{d}w \\
&= -\frac{\mathrm{i}}{2} \times \oint_{|w|=1} \frac{1}{w}\mathrm{e}^{w}\mathrm{d}w \\
&= -\frac{\mathrm{i}}{2} \times 2\pi\mathrm{i} \\
&= \pi.
\end{aligned}
$$

□

4

4.1

记

$$f(z)=\frac{\ln(z+3)}{z^2+5},$$

在 C 包围的单连通区域内，$f(z)$ 解析. 由 Cauchy 积分公式得

$$\begin{aligned}\oint_C \frac{\ln(z+3)}{z(z^2+5)}\mathrm{d}z &= \oint_C \frac{f(z)}{z-0}\mathrm{d}z\\ &= 2\pi\mathrm{i}\cdot f(0)\\ &= \mathrm{i}\frac{2\pi\ln 3}{5}.\end{aligned}$$

4.2

奇点 $(0,0)$ 在 C 包围的单连通区域内，记 $f(z)=\mathrm{e}^{-z}\sin(2z)$，则

$$f'(z)=(2\cos 2z-\sin 2z)\,\mathrm{e}^{-z}.$$

$f(z)$ 在 C 包围的单连通区域内解析，由导数 Cauchy 积分公式

$$f^{(n)}(z)=\frac{n!}{2\pi\mathrm{i}}\oint_C \frac{f(w)}{(w-z)^{n+1}}\mathrm{d}w,$$

得

$$f'(a)=\frac{1}{2\pi\mathrm{i}}\oint_C \frac{f(z)}{(z-a)^2}\mathrm{d}z.$$

令 $a=0$，得

$$\begin{aligned}\oint_C \frac{\mathrm{e}^{-z}\sin(2z)}{z^2}\mathrm{d}z &= \oint_C \frac{f(z)}{z^2}\mathrm{d}z\\ &= 2\pi\mathrm{i}f'(0)\\ &= 4\pi\mathrm{i}.\end{aligned}$$

4.3

奇点 $(0,1),(0,-1)$ 均在 C 包围的单连通区域内，用割线 $[-2,2]$ 将积分路径切为两个半圆 C_1,C_2. 记

$$f(z)=\frac{z^2-1}{z+i},\quad g(z)=\frac{z^2-1}{z-i},$$

$f(z)$ 在 C_1 内解析，$g(z)$ 在 C_2 内解析. 由 Cauchy 积分公式得

$$\begin{aligned}\oint_C \frac{z^2-1}{z^2+1}\mathrm{d}z &= \oint_{C_1} \frac{f(z)}{z-i}\mathrm{d}z+\oint_{C_2} \frac{g(z)}{z+i}\mathrm{d}z\\ &= 2\pi\mathrm{i}f(i)+2\pi\mathrm{i}g(-i)\\ &= 0.\end{aligned}$$

4.4

当 $n \in \mathbb{N}$ 时，$(z-a)^n$ 在 C 内解析，由 Cauchy 定理，

$$\oint_C (z-a)^n \,\mathrm{d}z = 0.$$

当 $n \in \mathbb{Z}^-/\{-1\}$ 时，奇点 $z=a$ 在 C 内，由导数 Cauchy 积分公式

$$f^{(n)}(z) = \frac{n!}{2\pi\mathrm{i}} \oint_C \frac{f(w)}{(w-z)^{n+1}} \mathrm{d}w,$$

故

$$\oint_C \frac{\mathrm{d}z}{(z-a)^{-n}} = \frac{2\pi\mathrm{i}}{(-n-1)!} f^{(-n-1)}(a),$$

其中 $f(z)=1$.

故 $f^{(-n-1)}(a)=0$，从而

$$\oint_C (z-a)^n \,\mathrm{d}z = 0.$$

若 $n=-1$，则

$$\begin{aligned}\oint_C \frac{\mathrm{d}z}{z-a} &= 2\pi\mathrm{i} f(a) \\ &= 2\pi\mathrm{i}.\end{aligned}$$

综上，

$$\oint_C (z-a)^n \,\mathrm{d}z = \begin{cases} 0, & n \neq -1, \\ 2\pi\mathrm{i}, & n = -1. \end{cases}$$

4.5

奇点 $(1,0),(2,0)$ 均在 C 内，记

$$f(z) = \frac{\sin(\pi z) + \cos(\pi z^2)}{z-1}, \quad g(z) = \frac{\sin(\pi z) + \cos(\pi z^2)}{z-2},$$

$f(z)$ 在 C_1 内解析，$g(z)$ 在 C_2 内解析，由 Cauchy 积分公式得

$$\begin{aligned}\oint_C \frac{\sin(\pi z) + \cos(\pi z^2)}{(z-1)(z-2)} \mathrm{d}z &= \oint_{C_1} \frac{f(z)}{z-2} \mathrm{d}z + \oint_{C_2} \frac{g(z)}{z-1} \mathrm{d}z \\ &= 2\pi\mathrm{i} f(2) + 2\pi\mathrm{i} g(1) \\ &= 2\pi\mathrm{i} + 2\pi\mathrm{i} \\ &= 4\pi\mathrm{i}.\end{aligned}$$

4.6

在 C 上 $z \neq 0$，且恒有 $|z| = 8$，从而

$$\begin{aligned}
\oint_C \frac{\overline{z}}{z^2 - \overline{z}} \mathrm{d}z &= \oint_C \frac{z\overline{z}}{z^3 - z\overline{z}} \mathrm{d}z \\
&= \oint_C \frac{|z|^2}{z^3 - |z|^2} \mathrm{d}z \\
&= \oint_C \frac{64}{z^3 - 64} \mathrm{d}z \\
&= 64 \times \oint_C \frac{1}{z^3 - 4^3} \mathrm{d}z.
\end{aligned}$$

C 内有 3 个奇点 $z = 4$，$4\mathrm{e}^{\mathrm{i}\frac{2}{3}\pi}$，$4\mathrm{e}^{\mathrm{i}\frac{4}{3}\pi}$.

用割线将 C 切为 3 个 $\frac{1}{3}$ 圆 C_1, C_2, C_3，记

$$f(z) = \frac{1}{\left(z - 4\mathrm{e}^{\mathrm{i}\frac{2}{3}\pi}\right)\left(z - 4\mathrm{e}^{\mathrm{i}\frac{4}{3}\pi}\right)}, \quad g(z) = \frac{1}{(z-4)\left(z - 4\mathrm{e}^{\mathrm{i}\frac{4}{3}\pi}\right)}, \quad h(z) = \frac{1}{(z-4)\left(z - 4\mathrm{e}^{\mathrm{i}\frac{2}{3}\pi}\right)},$$

则

$$\begin{aligned}
\oint_C \frac{\overline{z}}{z^2 - \overline{z}} \mathrm{d}z &= 64 \times \oint_C \frac{1}{z^3 - 4^3} \mathrm{d}z \\
&= 64 \times \left(\oint_{C_1} \frac{f(z)}{z-4} \mathrm{d}z + \oint_{C_2} \frac{g(z)}{z - 4\mathrm{e}^{\mathrm{i}\frac{2}{3}\pi}} \mathrm{d}z + \oint_{C_3} \frac{h(z)}{z - 4\mathrm{e}^{\mathrm{i}\frac{4}{3}\pi}} \mathrm{d}z \right) \\
&= 64 \times \left[2\pi\mathrm{i} f(4) + 2\pi\mathrm{i} g\left(4\mathrm{e}^{\mathrm{i}\frac{2}{3}\pi}\right) + 2\pi\mathrm{i} h\left(4\mathrm{e}^{\mathrm{i}\frac{4}{3}\pi}\right) \right] \\
&= 128\pi\mathrm{i} \left[f(4) + g\left(4\mathrm{e}^{\mathrm{i}\frac{2}{3}\pi}\right) + h\left(4\mathrm{e}^{\mathrm{i}\frac{4}{3}\pi}\right) \right] \\
&= 8\pi\mathrm{i} \times \left(3 - \frac{3}{2} - \frac{3\sqrt{3}}{2}\mathrm{i} - \frac{3}{2} + \frac{3\sqrt{3}}{2}\mathrm{i} \right) \\
&= 0.
\end{aligned}$$